Muyu He

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Education

Master of Science, Computer and Information Technology **Sept 2023 - May 2025**

University of Pennsylvania GPA 3.81/4.0

Bachelor of Arts, Philosophy **Sept 2018 - Dec 2021**

University of California, Los Angeles GPA 3.96/4.0

Work Experience

Research Scientist **July 2025 - Current**

Collinear.ai - Seed-stage AI startup specializing in LLM post-training and RL environments *Sunnyvale, US*

- Served as the solo contributor to *spider*, an API-based on-policy distillation framework
  - First engine to natively integrate MCP / software tool calls into on-policy distillation to support on-policy distillation across 100+ turns and 128K+ tokens
  - Unique cross-tokenizer alignment algorithm to enable arbitrary pairs of teacher-student distillation validated on models of scale 8B-685B
  - Adopted by the research community, with 95 GitHub stars and 100+ clones every two weeks
- Led a group of 4 research to run post-training data generation for a model (Apriel-1.5-15B-Thinker) to reach close-SOTA in competitive and terminal coding
  - Generated 5B tokens from three open-source models with multi-stage filtering strategy, lifting model performance from baseline (Nvidia OCR 58%) to close SOTA (71%) at the model size (14B).
  - Conducted 4 systematic ablations of data quantity (1K, 10K, 30K, 100K) on two models to validate a novel finding regarding the post-training scaling law of coding performance, ie. model performance first drops then recovers and surpasses
- Led the development of a production-grade chat API, *TraitBasis*, that dynamically steers model traits as part of the integrated evaluation pipeline of several labs (Sierra, Together.ai)
  - Achieved granular control of 4 traits (skeptic, impatient, etc.) for two model families (Llama3.1, Qwen3) that surpasses prompt- and training-based techniques in realism and controllability, as is judged pairwise by 3 human annotators on 20 user personas across 1.2K samples
  - Attained drastic data efficiency over traditional methods by extracting from only 4 conversation samples to outperform LoRA/full-finetune trained on 4K samples
  - Significantly improved the difficulty of prominent user-interaction benchmarks, in particular τ-bench where *TraitBasis* simulates a user with high fidelity, observing 5-10% drop across two domains (airline and retail) for three frontier agentic models (GPT-5, Kimi-K2, GLM-4.5)

Machine Learning Engineer Intern **Apr 2024 - July 2024**

SenseTime - Publicly-traded AI company pioneering in LLMs and computer vision *Shanghai, China*

- Led a team of 6 to train and deploy a production-grade GUI agent that reaches close-SOTA on Chinese mobile apps
  - Conducted SFT of VLMs of 2B, 7B, and 20B on 40+ real-world GUI environments including web browsing, music playing, and food ordering, observing superior performance against two open-source SOTAs (MobileAgent, AppAgent)
  - Optimized the distributed training pipeline on a cluster of 8+ nodes across three GPU types (Ampere, Hopper, Volta), employing specific key techniques for each GPU architecture including loss scaling, stage1-3 zero sharding, different forms of parallelism, etc.
  - Creatively implemented a synthetic data pipeline to create 10k high-quality screen navigation data samples, leveraging Android SDK to perform depth-first search and using the Dijkstra algorithm to synthesize optimal long-horizon plans spanning across 5-8 steps
  - Innovated a synthetic CPU-only data pipeline to create 6k home screen data samples with ground truth coordinates, using OpenCV to generate parametric layouts with 5+ layers that resemble real-world Android systems and scale up total data size by 5x

Publications

- **Muyu He***, Yuchen Liu*, Qingya Huang, Li Zhang *Equal contribution
  Do Value Vectors in Deep Layers Need Context from the Residual Stream?
  (Under review) Empirical Methods in Natural Language Processing (**EMNLP 2026**)

- **Muyu He**, Anand Kumar, Meghana Arakkal Rajeev, Soumyadeep Bakshi, James Zou, Nazneen Rajani
  Impatient Users Confuse AI Agents: High-fidelity Simulations of Human Traits for Testing Agents
  (Accepted by) Association for Computational Linguistics(**ACL Oral 2026**)

- Yuan Yuan*, **Muyu He***, Adil Shahid, Jiani Huang, Ziyang Li, Li Zhang *Equal contribution
  Evaluating Deductive Reasoning via Detective Games
  (Accepted by) Empirical Methods in Natural Language Processing (**EMNLP 2025**)

- Xingyu Fu*, **Muyu He***, Yujie Lu, William Yang Wang, Dan Roth *Equal contribution
  Commonsense-T2I Challenge: Can Text-to-Image Generation Models Understand Commonsense?
  (Accepted by) Conference On Language Modeling 2024 (**COLM 2024**)

Research

The Causal Mechanism for LLMs to Produce 'Attention Sinks' **Nov 2025 - present**
In collaboration with researchers from NUS and Drexel. GitHub (89 stars)

- Led 3 researchers to investigate the open problem of how LLMs produce dimension outliers only for the first token, testing on three models families (Qwen3, Llama3, Qwen3-Next) to reveal consistent novel findings across models
- Discovered the existence of a single dimension, the 'sink dimension', in high-dimension hidden space (4096 dims), to produce attention sinks, successfully ablating its importance on 20+ layers for a single model
- Found the existence of a registration mechanism whereas the model 'registers' the sink dimension prior to sink emergence and amplifies it across 5-6 layers before outputting an outlier that forms the sink

The Manifold learned by LLM to Drive Evil Behaviors **Aug 2025 - Sep 2025**
Independent. GitHub (122 stars)

- Creatively used only 3 conversation samples to extract an 'evil vector' in a frontier model with heavy alignment training (GPT-OSS-20B) to jailbreak 4 distinct misaligned scenarios
- Discovered the novel insight that the 'evil vector' suppresses key alignment terms in the token context by analyzing on sequences of 1K+ length
- Built a production-ready sparse autoencoder training pipeline to train on 10M GPT-OSS-20B tokens to find 256 alignment-related latents in high-dimension model space

Skills

Programming	Python, Java, C, C++, CUDA, PyTorch, numpy, SQL, Git
Deep learning	Large language models, pretraining, on-policy distillation, natural language processing, inference optimization, reinforcement learning, mechanistic interpretability